

Simulation parameters: Stability Analysis of Grid-Following and Grid-Forming Converters Connected to Generators

The key system parameters used in the small signal stability analysis of grid-following and grid-forming converters connected to synchronous generators are summarized in the following:

- **Synchronous generator:** $S_{SG} = 500$ MVA, $V_{SG}^{ref} = 24$ kV, $R_s = 0.003$ pu, $L_s = 0.15$ pu, $L_m^d = 1.6598$ pu, $L_m^q = 1.6099$ pu, $L_{k1}^q = 0.7252$ pu, $R_{k1}^q = 0.00619$ pu, $L_{k2}^q = 0.125$ pu, $R_{k2}^q = 0.02368$ pu, $L_k^d = 0.1713$ pu, $R_k^d = 0.0284$ pu, $L_f^d = 1.8245$ pu, $R_f^d = 5.998 \cdot 10^{-4}$ pu.
- **Exciter:** $\tau_C = 1$ s, $\tau_B = 10$ s, $k_A = 200$, $\tau_A = 0.015$ s.
- **Governor & Turbine:** $K_{HP} = 0.25$, $K_{LP} = 0.75$, $T_{RH} = 5$ s, $k = 5\%$, $\tau_R = 0.1$ s.
- **Voltage source converter** (single converter, then the aggregate model of N parallel-connected converters could be derived): $S_{VSC} = 5$ MVA, $N = [50, 100, 150]$, $V_{VSC} = 660$ V, $V_{DC} = 1500$ V, $C_c = 5000$ μ F, $L_f = 0.25$ pu, $R_f = 0.1148$ pu, $C_f = 0.05$ pu, $R_{fc} = 0.3053$ pu, $P_{ref} = 0.9$ pu, $Q_{ref} = 0.1$ pu.
- **Grid following control:** $k_i^{PLL} = 40$ rad/(Vs), $k_p^{PLL} = 0.4$ rad/(Vs²), $\tau_i = 0.5$ ms, $\tau_{DC} = 5.9$ ms.
- **Grid forming control:** $\tau_i = 0.5$ ms, $\tau_v = 5$ ms, $\zeta_v = 0.707$, $m_p = 5\%$, $n_q = 2\%$.
- **Transformer** (single transformer, then the aggregate model could be derived): $S_{tr} = 5$ MVA, $X_{tr} = 0.05$ pu.
- **Grid:** $f_n = 50$ Hz, SCR = 3.5–1.5, $r_l = 0.1$ Ω /km, $l_l = 1$ mH/km.